THE DETERMINANT OF A MATRIX

LINEAR ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

The Determinant of a Matrix

Minors and Cofactors

Laplace Expansion

THE DETERMINANT OF A MATRIX

The determinant of a 1×1 matrix

$[\boldsymbol{a}]$ is **a**

Find the **determinant** of the matrix

$[\mathbf{5}]$

$$\boldsymbol{det}([\mathbf{5}]) = \mathbf{5}$$

Find the **determinant** of the matrix

$$\begin{bmatrix} 3 & -2 \\ 4 & 5 \end{bmatrix}$$

The determinant of a 2×2 matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is $\boldsymbol{ad - bc}$

$$det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

The determinant of a 3×3 matrix

$$\mathbf{A} = \begin{array}{c} \begin{array}{ccc} + & - & + \end{array} \\ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \end{array}$$

where,

$$a_{ij} = entry$$

$$M_{ij} = minor$$

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

$$\boldsymbol{M_{11}} = \begin{vmatrix} \boldsymbol{a_{22}} & \boldsymbol{a_{23}} \\ \boldsymbol{a_{32}} & \boldsymbol{a_{33}} \end{vmatrix}$$

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} \boldsymbol{a_{11}} & \boldsymbol{a_{12}} & \boldsymbol{a_{13}} \\ \boldsymbol{a_{21}} & \boldsymbol{a_{22}} & \boldsymbol{a_{23}} \\ \boldsymbol{a_{31}} & \boldsymbol{a_{32}} & \boldsymbol{a_{33}} \end{bmatrix}$$

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Determine the ij minors $\boldsymbol{M_{11}, M_{12}, M_{13}}$ of the matrix

$$\mathbf{A} = \begin{bmatrix} \mathbf{2} & \mathbf{5} & \mathbf{3} \\ \mathbf{1} & \mathbf{-4} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} & \mathbf{-1} \end{bmatrix}$$

Find the **determinant** of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix}$$

then the **cofactor** of element a_{ij} denoted by $\boldsymbol{C_{ij}}$ is

$$\boldsymbol{C_{ij} = (-1)^{i+j} M_{ij}}$$

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Cofactor of A,

$$\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix}$$

If A is an $n \times n$ matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & a_{nn} \end{bmatrix}$$

then the determinant of A,

$$det(A) = a_{11}M_{11} - a_{12}M_{12} \ldots \pm a_{nn}M_{nn}$$

Find the **determinant** of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

LAPLACE EXPANSION

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$det(A) = (a_{11}a_{22}a_{33}) + (a_{12}a_{23}a_{31}) + (a_{13}a_{21}a_{32})$$

$$-(a_{31}a_{22}a_{23}) - (a_{32}a_{23}a_{11}) - (a_{33}a_{21}a_{12})$$

Find the **determinant** of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix}$$

4X4 MATRIX

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$det(A) = (a_{11}a_{22}a_{33}a_{44}) + (a_{12}a_{23}a_{34}a_{41}) + (a_{13}a_{24}a_{31}a_{42}) + (a_{14}a_{21}a_{32}a_{43})$$

$$-(a_{41}a_{32}a_{23}a_{14}) - (a_{42}a_{33}a_{24}a_{11}) - (a_{43}a_{34}a_{21}a_{12}) - (a_{44}a_{31}a_{22}a_{13})$$

Find the **determinant** of the matrix

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix} \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}$$

$$det(A) = (a_{11} \cdots) \cdots + (a_{1n} \cdots)$$

$$-(a_{n1} \cdots) \cdots - (a_{nn} \cdots)$$

SEATWORK